Tutorial 12

Exercise 1: Decrementing counters

1. Let us add a decrement operation `DEC` to the k-bit counter example from the lecture, which is implemented analogously to the increment operation. In particular, the cost of one decrement is the number of bits inspected and/or updated.

   Show that a sequence of n operations on an initially zero counter can cost at least $n \cdot k$ time.

2. Let us add a reset operation `RESET` to the k-bit counter example from the lecture (i.e., without decrement). Again, the cost of one reset is the number of bits inspected and/or updated.

   Implement a counter as an array of bits so that any sequence of n increments and resets takes $\mathcal{O}(n)$ time on an initially zero counter. Show your implementation of the operations and the analysis.

   Hint: Keep a pointer to the highest-order 1.

Exercise 2: Applying amortized analysis

Consider a sequence of n operations on some data structure in which the i-th operation costs i if i is an exact power of two (i.e, $i \in \{1, 2, 4, 8, 16, \ldots\}$) and 1 otherwise.

1. Use aggregate analysis to determine the amortized cost per operation of such a sequence (for an arbitrary n).
2. Use the accounting method to determine the amortized cost per operation of such a sequence (for an arbitrary n).
3. Challenge: Use the potential method to determine the amortized cost per operation of such a sequence (for an arbitrary n).

Exercise 3: Bounded stacks

You perform a sequence of `PUSH` and `POP` operations on a stack whose size never exceeds k. After every k operations, a copy of the entire stack is made automatically, for backup purposes.

Show that the cost of n stack operations, including the copying of the stack, is $\mathcal{O}(n)$ by assigning suitable amortized costs to the various stack operations.

Exercise 4: Potential functions

Suppose you have a potential function Φ such that $\Phi(D_i) \geq \Phi(D_0)$ for all i, but $\Phi(D_0) \neq 0$. Show that there is a potential function Φ' such that $\Phi'(D_0) = 0$, $\Phi'(D_i) \geq 0$ for all i, and the amortized costs using Φ' are the same as the amortized costs using Φ.